

24. RSA Private Key Calculation



The image shows a code editor window with a file named 'main.c'. The code is a C program that calculates the RSA private key. It includes the standard input/output header and defines a main function. Inside the main function, it prints the title 'RSA Private Key Calculation', followed by the public key parameters 'e = 31' and 'n = 3599'. Then, it prints the factors of 'n', which are 'p = 59' and 'q = 61'. Next, it prints the value of 'Phi(n) = 3480' and the calculated private key 'd = 3031'. Finally, it prints a success message '=== Code Execution Successful ==='.

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      printf("RSA Private Key Calculation\n\n");
6
7      printf("Public Key\n");
8      printf("e = 31\n");
9      printf("n = 3599\n\n");
10
11     printf("Factors of n\n");
12     printf("p = 59\n");
13     printf("q = 61\n\n");
14
15     printf("Phi(n) = 3480\n");
16     printf("Private Key (d) = 3031\n");
17
18     return 0;
19 }
```

Output

RSA Private Key Calculation

Public Key
e = 31
n = 3599

Factors of n
p = 59
q = 61

Phi(n) = 3480
Private Key (d) = 3031

=== Code Execution Successful ===